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Exploring evolution in the context of molecular genetics and ecology: a dual perspective

Merav Siani^{1,2*} and Anat Yarden¹

Abstract

The teaching of evolution stands as a cornerstone in the realm of biological sciences, yet how best to frame and teach the complex web of concepts that are a part of evolutionary theory is still under debate. To address this issue, we propose two sequences for teaching the evolution ideas and concepts that are included in the Israeli curriculum for upper secondary school, starting from either the foundational principles of molecular genetics or the intricate dynamics of ecology or integrating both. This approach involves considering the strengths of both molecular genetics and ecology as frameworks for understanding evolution, recognizing that each perspective offers valuable insights that can enrich students' understanding of the topic. Molecular genetics is the area of evolutionary theory that relies on terms such as genes, alleles, and mutations. Ecology offers a broader, more holistic view of evolution and includes the dynamic interplay between organisms and their environment. The molecular genetics sequence focuses on the mechanism of evolution and the ecology sequence focuses on the external factors that affect the mechanism. This dual approach creates options for teachers; they can take into consideration each path's advantages and use the characteristics of their classes to choose one of the suggested perspectives or integrate both perspectives to teach evolution.

Keywords Teaching sequence, Molecular genetics, Ecology, Dual perspective

Background

The teaching of evolution stands as a cornerstone in the realm of biological sciences, yet the approach to its instruction often triggers lively debates among educators, scholars, and policymakers (Hanisch and Eirdosh 2020; Siani and Yarden 2020). Central to this debate is the question of context—how best to frame and present the complex web of concepts that are a part of evolutionary theory (Christensen and Lombardi 2020). To address this debate, we suggest two sequences for teaching

evolution in upper secondary school: starting from the foundational principles of molecular genetics or from the intricate dynamics of ecology. We delve into these two distinct contextual lenses by considering the strengths of each as frameworks for understanding evolution, and recognizing that each perspective offers valuable insights that can enrich students' understanding of the topic.

The rationale of this dual perspective is twofold. First, molecular genetics serves as the molecular basis of evolutionary theory. The mechanisms of evolution are based on genes, alleles, and mutations. By exploring evolution through the lens of molecular genetics, we unravel the details of inheritance, adaptation, and natural selection, providing students with a solid foundation for possible evolutionary mechanisms that are rooted in molecular genetics. Second, ecology offers a broader, more holistic

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view of evolution—one that extends beyond the borders of the genome, and includes the dynamic interplay between organisms and their environment. In considering the balance between ecosystems and biodiversity, the ecological perspective enriches students' understanding of evolution by contextualizing the changes within the broader tapestry of life on earth.

In this study, we navigate the complex field of evolution, exploring the nuances of molecular genetics and ecology and the invaluable contributions of teaching evolution from each perspective as well as the integration of both. This exploration attempts to highlight and inspire a deeper appreciation for the sequence of teaching evolution. Thus, teachers can tailor their teaching methods to best meet the needs of their students, drawing on the strengths of both genetics and ecology to provide a comprehensive and engaging learning experience.

Central ideas in learning evolution

The concepts taught in schools as part of the topic of evolution are often referred to as threshold concepts, meaning that without understanding or interpreting them, the learner cannot understand natural selection (Meyer and Land 2003). According to Ross et al. (2010), understanding evolution involves three main mechanisms that make up this theory's framework: (i) the physical and genetic variation that exists in a population, including the concepts of probability and randomness; (ii) variation in the features of an organism that are inherited as separate units; and (iii) a time scale that involves millions of years. According to Ross (2010), these are considered threshold concepts because they are integrative, transformative, and once they are acquired, the student can develop a deeper understanding of evolution (Ross et al. 2010).

Zabel and Gropengiesser (2011) refer to five key ideas of evolutionary theory that are essential for learning to advance: (i) explaining, instead of merely describing the evolutionary changes; (ii) understanding that evolution occurs over several generations and does not refer to the "evolution" of an individual; (iii) explaining evolutionary changes by causal mechanisms rather than teleological reasons; (iv) understanding that natural selection is the main mechanism driving evolution; (v) understanding that evolution proceeds through variation and natural selection. If a student understands these key ideas, it implies that he or she understands how individuals differ from each other, which is essential for the explanation of evolutionary theory.

Tibell and Harms (2017) agreed with the threshold concepts proposed by Ross et al. (2010) and added that a full understanding of the theory of evolution requires the ability to understand two dimensions: (i) the threshold concepts of randomness, probability (Fiedler et al. 2017),

temporal scale, and spatial scale; and (ii) the key concepts of variation, heredity, and natural selection (Tibell and Harms 2017). Meaning, in addition to the threshold concepts proposed by Ross et al. (2010), concepts of variation and heredity are needed to learn about evolution and understand it.

The intertwining of key and threshold concepts was referred to by Göransson et al. (2020), stating that students should understand that the evolutionary process begins with random mutations that lead to phenotypic variation. This variation, combined with changes in the environment, can result in differential survival and reproduction. If this genetic variation is inherited, it can bring about a change in the population. Over longer periods, these changes can eventually generate new species (Göransson et al. 2020).

All of these concepts present the topics that are fundamental to teaching evolution at the high-school level, and there is no significant difference among researchers in defining these fundamental topics, which are also agreed upon by the writers of the Next Generation Science Standards (NGSS) (NRC 2012).

Challenges in learning evolution

One of the challenges in learning evolution is that students possess alternative conceptions about the subject (Harms and Reiss 2019; Siani and Yarden 2022). One extensively explored alternative conception relates to natural selection (Lucero et al. 2019; Prinou et al. 2008). When learning evolution, students often struggle to grasp that mechanisms in biological systems result from natural processes, such as selection, rather than intentional or need-based fulfillment (Kampourakis 2020). Natural selection is not the only problematic aspect when learning evolution; the comprehension of fundamental evolutionary concepts also remains insufficient. Students hold alternative conceptions about what constitutes evolution in biology, the primary mechanism of evolutionary changes, the actual scope of the theory of evolution, and the scientific meaning of the term "theory" (Prinou et al. 2008). Furthermore, when explaining evolutionary changes, students frequently resort to teleological and occasionally Lamarckian arguments (Eder et al. 2018).

In the United States (Hermann 2013) as well as in Israel (Siani and Yarden 2022), educators acknowledge the challenge posed by evolution for students due to difficulties in grasping the dimension of time. Students harbor fixed ideas that conflict with scientific perspectives, such as species change and extended time spans. To understand these notions, which may go against students' prior experiences, background knowledge, and perhaps personal beliefs, a conceptual change might be necessary (Heddy and Sinatra 2013).

The complexities of teaching evolution contribute to the precarious position of this topic in science curricula worldwide. Recent findings highlight insufficient acknowledgment of, or emphasis on the significance of understanding evolution in school curricula across Europe (Mavrikaki et al. 2024). They also indicate a failure to follow the recommendations of educational research organizations such as the National Research Council (NRC) (NRC 2012) which advocate for the comprehensive inclusion and promotion of evolution teaching throughout compulsory education. Notably, data from recent research reveal that most of the curricula in Europe incorporate less than half of the essential learning objectives crucial for fostering scientific literacy in evolution (Mavrikaki et al. 2024).

Learning the topic of evolution around the world

While examining the place of evolution ideas in standards across the globe, nine high-school biology curricula were reviewed (Zer-Kavod 2018), in Australia, England, New Zealand, Singapore, Scotland, Finland, the Canadian Province of British Columbia, and Virginia and California in the United States. The topic of evolution was found to be part of all of the examined curricula. Evolution was taught as a separate topic in the high-school biology curricula in Virginia, California (following the NGSS), New Zealand, Finland, Singapore, and the Canadian Province of British Columbia. In the four other locations, evolution was part of another topic in the curriculum: genetics in Australia, England and Scotland, and ecology in Austria (Zer-Kavod 2018).

In Israel, evolution has been placed more centrally in the core high-school biology curriculum since 2016, in the context of ecology (Siani and Yarden 2020). It is now mentioned in the curriculum in two contexts in the section referring to ecosystems: “Evolutionary processes affect the prevalence of traits that characterize the species, and the diversity of the species” and “Man influences the process of evolution of species” (Israeli Ministry of Education. *Biologia* 2016). In addition to the concepts dealing with evolution that are detailed in the curriculum, it should be noted that the Israeli high-school biology curriculum is based on seven central ideas in biology that serve as the curriculum’s organizational foundations. One of these central ideas is *The theory of evolution* (Israeli Ministry of Education 2016). Thus, even though the topic of evolution is placed in the Israeli curriculum in the context of ecology, as can be understood from this central idea, ecology and genetics are both important for understanding the central concepts in evolution. Yet, teaching evolution only in the context of ecology without going into the molecular genetics processes leading to randomness and probability, might lead to difficulty in

understanding the mechanisms of natural selection and what drives the processes of evolution. For that reason, in this paper we look into the option of teaching evolution as a sequence in the context of ecology and as a sequence in the context of molecular genetics as well as integrating both of them, pointing out the advantages of each.

Learning evolution in the context of molecular genetics

Because in some countries, evolution is taught in the context of molecular genetics (Zer-Kavod 2018), we address the notion that learning evolution in the context of molecular genetics improves understanding (Elmesky 2013) and acceptance (Mead et al. 2017) of the theory. According to Smith et al. (Smith et al. 2009), it is important to teach evolutionary biology after understanding the universality of the genetic evidence of evolution, which is crucial for understanding evolutionary theory. This claim suggests that evolution should be taught in ways that integrate molecular genetics ideas and evidence (Smith et al. 2009).

When integrating the teaching of evolution with molecular genetics, students might better understand how genetic changes at the molecular level contribute to evolutionary processes (Hill et al. 2021). Molecular genetics reinforces the evidence for a common ancestry among different species. Comparative genomics and the molecular homologies that are shared among organisms provide compelling evidence for the idea of a shared evolutionary history (Koenen et al. 2020). In addition, molecular genetics helps clarify the mechanisms driving evolution, such as genetic drift, natural selection, and gene flow. Students can even understand how changes in DNA sequences, such as epigenetic changes, might lead to phenotypic variations and contribute to the overall diversity of life (Sarkies 2020; Ashe et al. 2021).

Moreover, learning evolution in the context of genetics may also alleviate some of the controversy surrounding the topic of evolution (Siani and Yarden 2020), because variation that leads to evolution occurs in the genetic material, and there is less controversy regarding genetic concepts (Donovan et al. 2020). Thus, teaching evolution in the context of molecular genetics might lead to higher acceptance of evolution, and initial empirical evidence suggests that this may be so. Mead et al. (2017) compared teaching molecular genetics before evolution to teaching evolution before genetics among United Kingdom high-school students. They found that teaching evolution after genetics improved evolution understanding by 5–10% and increased acceptance of the theory among the students, who showed long-term retention. Another study compared 2269 students, approximately half of them studying molecular genetics before evolution. These

teachers reported that there was a clear advantage when basing evolution in molecular genetics for understanding the mechanics of evolution. Other teachers, who did not teach evolution with molecular genetics, described difficulties in teaching evolution without the molecular genetics background (Drits-Esser et al. 2021).

Furthermore, teaching evolution in the context of molecular genetics may encourage critical thinking skills. Teaching heredity and evolution in an integrated unit may enable students to analyze and interpret molecular data, make connections between genetic changes and evolutionary outcomes, evaluate scientific evidence, identify claims, and reason in scientific arguments (Homburger et al. 2019). High-school students in Spain were taught genetics prior to evolution as a way to build their understanding of fundamental concepts of genetics in combination with argumentation practices; they then applied those concepts to learning about evolution (Ageitos et al. 2019). Along with critical thinking skills, curiosity and interest may be motivated when learning genetics before evolution (Alonso and Palomares 2021), both of which are known to be important for students' motivation to learn a certain topic (Jirout 2020).

Thus, teaching evolution in the context of molecular genetics may enhance the understanding and acceptance of evolutionary principles, and lead to students' acquisition of scientific skills and motivation for learning.

Learning evolution in the context of ecology

Because in some countries, including Israel, evolution is taught in the context of ecology (Zer-Kavod 2018), we address the notion that learning evolution in the context of ecological systems improves understanding of evolutionary theory. One of the advantages of integrating evolution with ecology, might be an enhanced understanding for the students of how different species have evolved and adapted to their specific ecological niches, contributing to the overall biodiversity of ecosystems (Nesimyan-Agadi and Ben-Zvi 2023, 2022). In addition, evolutionary concepts are crucial for understanding population dynamics within ecosystems (Fisher and Pruitt 2020). In this context, students can explore how variation, adaptations, and natural selection influence the abundance and distribution of species in a given environment.

Moreover, teaching evolution in the context of ecology may provide insights into the diversity of populations, the impact of environmental changes on species survival, and strategies for preserving biodiversity (Brunner et al. 2019). According to Brodersen & Seehausen (Brodersen and Seehausen 2014), it is important to incorporate evolutionary processes into biodiversity-monitoring efforts

through enhanced collaboration between conservation practitioners, ecologists, and evolutionary biologists.

Furthermore, ecology and evolution are interconnected when considering how species adapt to changing environmental conditions. Understanding evolutionary processes may help students understand natural selection, showing the student how organisms evolve traits that enhance their survival and reproduction in response to environmental challenges (Sá-Pinto et al. 2021). Moreover, evolutionary insights may contribute to understanding the roles that different species play in providing ecosystem services. This knowledge is crucial for making informed decisions about conservation, land use, and sustainable resource management (Garant 2020). Evolutionary perspectives may also help students examine the impact of human activities on ecosystems. They can explore how human-induced changes, such as habitat destruction and climate change, affect the evolutionary trajectories of species and ecosystems (Pivello et al. 2021).

Referring to scientific skills, studying the interactions between evolution and ecology may encourage students to engage in scientific inquiry. They can observe and analyze ecological patterns, collect data, and draw conclusions about the evolutionary forces shaping the dynamics of ecosystems (Nesimyan-Agadi and Ben-Zvi 2022). Taken together, teaching evolution within the context of ecology may provide a comprehensive and integrated understanding of the natural world. It may enable connecting evolutionary principles with ecological dynamics, enriching students' perspectives on the interplay between life's diversity and the environments in which it thrives.

How do teachers teach evolution?

Teachers do not always feel obligated to follow the biology curriculum that is published by the authorities, i.e., not all of the teachers teach in the suggested sequence. In Israel, for example, even though evolution is part of the ecology curriculum, teachers have claimed that they do not always teach according to the suggested curriculum. In a recently conducted study among Israeli middle-school and high-school science teachers, we found that only 37.14% teach evolution throughout the biology curriculum, while 35.71% teach evolution as part of ecology and 27.14% teach it as part of genetics (Siani and Yarden, 2022, unpublished data). In other words, the curriculum is only a suggestion for some of these teachers. Because teachers do not consider the curriculum to be a rigid and absolute protocol, we recognized that a teaching sequence that shows the two contexts in which evolution could be taught might be useful for teachers and teacher trainers.

Integrating the teaching of evolution with the context of genetics and ecology

To address the question of the two contexts in which evolution can be taught, we suggest a teaching sequence for evolution. Breaking down the subject matter into a structured sequence could potentially make it easier for students to understand and for teachers to teach, might be more engaging for students, and might be better aligned with educational objectives, such as enhancing scientific literacy.

To integrate the two topics—genetics and ecology—with evolution, we incorporated all of the evolutionary concepts that are taught in the Israeli high-school biology curriculum into a concept map (Fig. 1). This map visually integrates the key topics of genetics and ecology within the framework of evolution. The map highlights the interconnections between the domains of molecular genetics, ecology, and evolution, demonstrating how they collectively contribute to our understanding of evolutionary processes. The core of the concept map shows the three main ideas of evolution, as defined in the Israeli

curriculum (Fig. 1, numbered and colored in yellow), and the concepts of evolution (Fig. 1, blue rectangles with different frames). Learning molecular genetics is suggested for understanding some of these concepts (Fig. 1, blue rectangles with orange frame). For understanding other concepts, learning evolution in the context of ecology is suggested (Fig. 1, blue rectangles with green frame). For understanding additional concepts, learning evolution in the context of both ecology and molecular genetics is suggested (Fig. 1, blue rectangles with orange and green frame). In addition, the concept map shows the basic concepts of molecular genetics that are taught as part of the topic *The Cell* and can contribute to learning evolution (Fig. 1, orange rectangles), as well as the basic concepts of the topic *Ecology* that are connected to evolution (Fig. 1, green rectangles). Learning molecular genetics provides a foundational understanding of how genetic variation arises through mutations, meiosis, and sexual reproduction. This genetic variation is the raw material for evolution. Ecology provides the environmental context in which natural selection operates. Biotic and

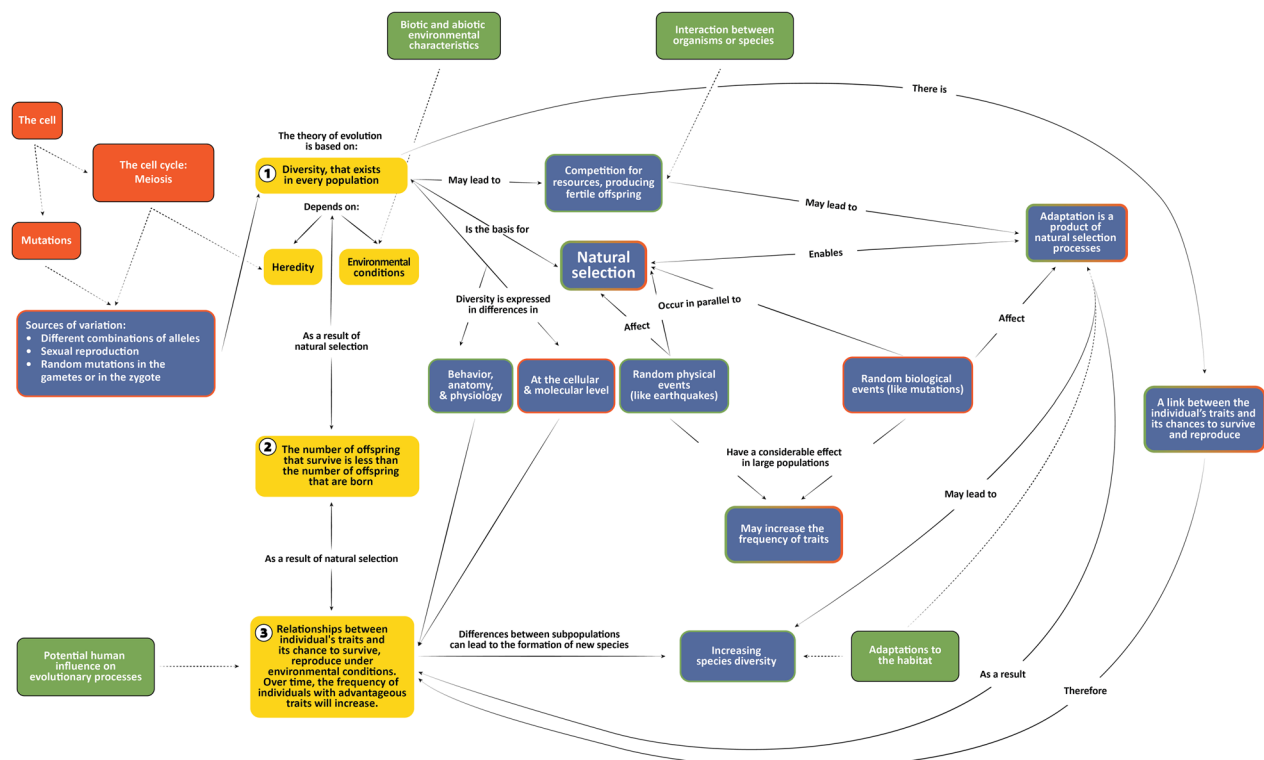


Fig. 1 The concept map for teaching evolution in the context of genetics and ecology. Orange rectangles: molecular genetics concepts that are taught as part of the topic *The Cell*. Green rectangles: ecology concepts that are taught as part of the topic *Ecology*. Yellow rectangles: main ideas of evolutionary theory. Blue rectangles with orange frame: concepts in the evolution curriculum for which learning molecular genetics first is suggested. Blue rectangles with green frame: concepts in the evolution curriculum for which learning ecology first is suggested. Blue rectangles with orange and green frame: concepts in the evolution curriculum for which learning ecology as well as molecular genetics first is suggested. Dashed lines: connections between molecular genetics concepts/ecology concepts and the concepts and ideas of evolution. Solid lines: connections between concepts and ideas in the evolution curriculum

abiotic environmental factors influence which traits are advantageous. Interactions between organisms and their environment, including competition for resources, shape evolutionary outcomes.

Overall, the concept map effectively bridges and integrates genetics and ecology, showing their combined role in driving evolution. It underscores the importance of understanding genetic variation and environmental influences to fully grasp the mechanisms of natural selection and adaptation. This integrated approach provides a comprehensive framework for students to understand the complexity of evolutionary biology.

To use the concept map (Fig. 1) easily as a teaching sequence, we propose two teaching sequences (Fig. 2), one in the context of molecular genetics (Fig. 2A) and the other in the context of ecology (Fig. 2B). We shall deal with each sequence on its own first and then deal with the integration of both contexts. Note that the

color-coding in Figs. 1 and 2 relates to the same ideas and concepts. For example, the yellow rectangles in both figures indicate the three main ideas upon which evolutionary theory is based.

Looking at Fig. 2A, diving into evolutionary theory might start with learning the cell cycle, including the terms meiosis and mutations. Both increase the variations stemming from several sources, which are the basic processes for diversity—a basic idea in evolutionary theory. Diversity depends on heredity and is expressed in differences at the cellular and molecular levels. Thus, learning molecular genetics may contribute to a thorough understanding of these concepts. Diversity is the basis for natural selection, one outcome of which is adaptation. The frequency of traits leading to adaptation is affected by natural selection, but also by mutations; once again, understanding this concept may help at this point as well. Individuals' traits are thus the basis for survival

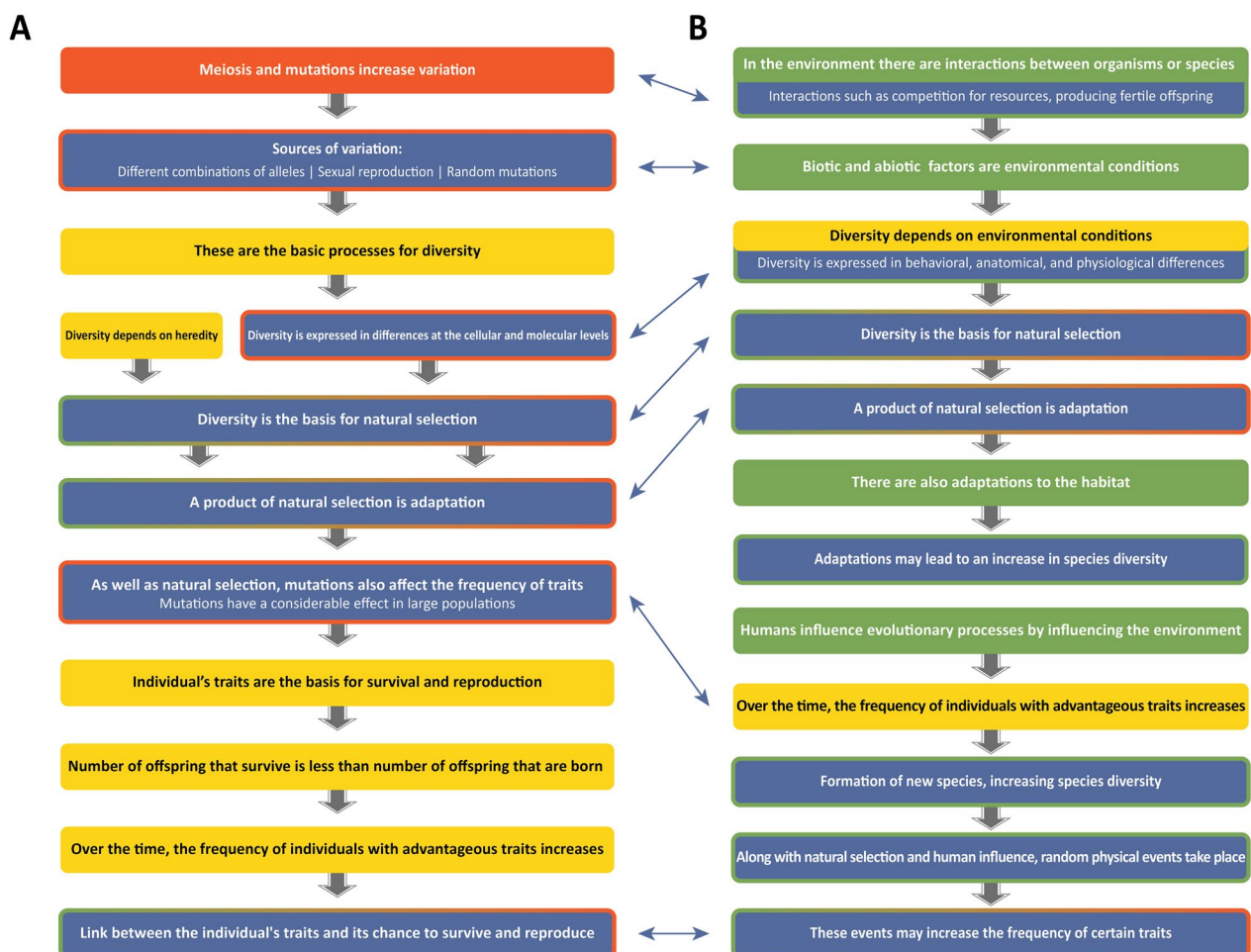


Fig. 2 Sequences for teaching evolution: **A** in the context of molecular genetics, and **(B)** in the context of ecology. The double-sided arrows show the possible integration of teaching evolution in both contexts. See Fig. 1 caption for color-coding key. The colors in both figures represent the same ideas and concepts

and reproduction, leading to two ideas in evolutionary theory: that the number of surviving offspring is less than the number that are born, and that the frequency of individuals with advantageous traits increases over time. This leads to the concept in evolution of a link between the individual's traits and its chances of surviving and reproducing, a concept that is actually connected to all of the aforementioned concepts and ideas.

Now we consider the sequence of learning evolution in the context of ecology (Fig. 2B). Figure 1 shows four points of intersection between the ecology and evolution curricula. Teaching evolution in the context of ecology may start with the ecological concept that there are several types of interactions between organisms or species in the environment, e.g., competition for resources and producing fertile offspring. Biotic and abiotic conditions are also part of the environment; these are taught in the ecology curriculum, and are important when introducing diversity that depends on environmental conditions. Diversity is a central idea in evolution that leads to the central concept of natural selection. An additional intersection of ecology with evolution is via adaptations to the habitat, which may lead to an increase in species diversity. Finally, the human influence on evolution represents another intersection between ecology and evolution curricula, examples being resistance to drugs and pesticides, and industrial melanism. These influences may lead to an increase in the frequency of advantageous traits, and thus the formation of new species or an increase in species diversity and, in the long run, an increase in the frequency of certain traits. Accordingly, learning evolution after learning one of the four ecology concepts (green rectangles in Fig. 2B) may help students better understand both ecology and evolution.

The integration of teaching molecular genetics and ecology shows their interplay when teaching key evolutionary processes. As represented in Fig. 2 by the double-sided arrows, a few concepts are common to both contexts: 'diversity', 'natural selection', 'survival' and 'adaptation'. These concepts enable to integrate between the two contexts and to move back-and-forth between the levels of organization that each topic deals with. Teaching about mutations and allelic variation as sources of diversity (genetics) on the cellular organization level, can easily go into discussions about how these variations might impact survival and reproduction producing fertile offspring on the population organization level (ecology). Additionally, students can explore how random mutations in the DNA of a moth lead to darker pigmentation, which then provides an advantage in a polluted, darker habitat. This is a classic case of industrial melanism that may integrate between molecular genetics and ecology. Another example is in the field of antibiotic resistance.

Molecular genetics enables to explain how specific genetic mutations, dealing with the cellular organization level, can lead to a variety of bacteria that some of them are antibiotics resistant, while ecology can provide the context of the overuse of antibiotics, that allows resistant bacteria to proliferate in certain environmental conditions, dealing with the ecosystem organization level. These real-world cases highlight the idea that neither genetics nor ecology can provide a full understanding of evolutionary change on their own. Rather, it is the back-and-forth between these domains that allows to capture the complexity and interdisciplinarity of evolutionary processes, presenting evolution as a dynamic interaction moving between the different levels of organization that each of the topics deals with.

Conclusions

In this paper, we discuss the advantages of teaching evolution in the context of molecular genetics and in the context of ecology as well as the integration of both, in order to enhance understanding of evolution, and minimize opposition to evolution. The concept map shows the basic concepts of molecular genetics that are taught as part of the topic *The Cell* which can lead to learning evolution, as well as basic concepts of the topic *Ecology* that are connected to evolution.

As can be concluded from the proposed teaching sequences, molecular genetics provides a foundational understanding of heredity and molecular mechanisms. This understanding helps students appreciate how natural selection acts on genetic variation to drive evolutionary change. Thus, to improve understanding of genetics and evolution, as well as to enhance evolution acceptance, teachers might focus more on the teaching of the genetic concepts that support the mechanisms of evolutionary change (Mead et al. 2017). This knowledge forms the basis for understanding evolutionary processes and genetic variations within populations. Evolution might be introduced after building upon the concepts of genetic variation and inheritance.

Ecology might also be introduced before evolution because it allows students to explore how organisms interact with their environments and how ecological factors influence evolutionary processes (Campbell 2018). In this way, ecological concepts such as adaptation, interactions, and ecosystem functioning might be taught prior to learning the wider concepts of evolution. In addition, learning along a sequence aligns with the hierarchical nature of biological concepts and facilitates a deeper understanding and integration of topics over time.

In fact, it may be concluded that the molecular genetics sequence focuses on the mechanism of evolution and the ecology sequence focuses on the external factors that

affect the mechanism. Genetic variations and mutations drive changes within populations, while the environment, including interactions with other organisms and human influences, determine which of these genetic variations will be advantageous. These external factors influence the direction, pace, and outcome of evolutionary processes. When these two sequences are combined, they provide a comprehensive understanding of evolution. Thus, teaching evolution from both perspectives presents the topic of evolution as a whole.

This dual approach, emphasizing the potential benefits of each of the perspectives, creates options for teachers; taking into consideration the advantages of both options and using the characteristics of their classes, so they can choose which suggested perspective to follow.

Limitations

The first limitation of this paper is that it is a theoretical paper. The suggested sequences, molecular genetics-based and ecology-based, are presented as standalone approaches with opportunities for integration between them, but the practical challenges of implementing an integrated framework have not been examined in classes. Integrating the frameworks in classes may be challenging since teachers' expertise in both genetics and ecology is different, there are curriculum constraints, and student have differences in learning backgrounds. All these might affect the probability and effectiveness to integrate these two contexts in teaching evolution.

Second, while the concept map (Fig. 1) provides a visual representation of the interconnectedness of genetics and ecology, it may not fully capture the complex nature of evolutionary processes. As such, the concept map, while useful as an instructional tool, should be seen as a starting point rather than a comprehensive model of evolutionary processes.

Third, this study focuses primarily on the Israeli curriculum, which may limit its' generalizability to other educational systems with a different curriculum and different standards and emphases on evolution education. Educational systems that prioritize molecular genetics might find the ecology-based sequence less intuitive or practical, and vice versa. Despite this limitation, it may be inspiring for teachers to teach according to the suggested concept map and perceive the consequences for students' learning.

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Author contributions

M.S. developed the teaching sequences after fruitful discussions and a thorough analysis of the evolution curriculum with A.Y. M.S. wrote the manuscript,

while A.Y. provided significant guidance throughout the writing process, revised, and edited all drafts of the manuscript.

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Competing interests

The authors declare no competing interests.

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